Algorithms

Lecture # 01

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Preliminaries

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- *Course*: Design and Analysis of Algorithms
- *Prerequisites*: Data Structures, Discrete Mathematics
- *Textbook:* Introduction to Algorithms,Cormen,Leiserson, Rivest, Stein, 2nd edition, 2001 The MIT Press.

Preliminaries

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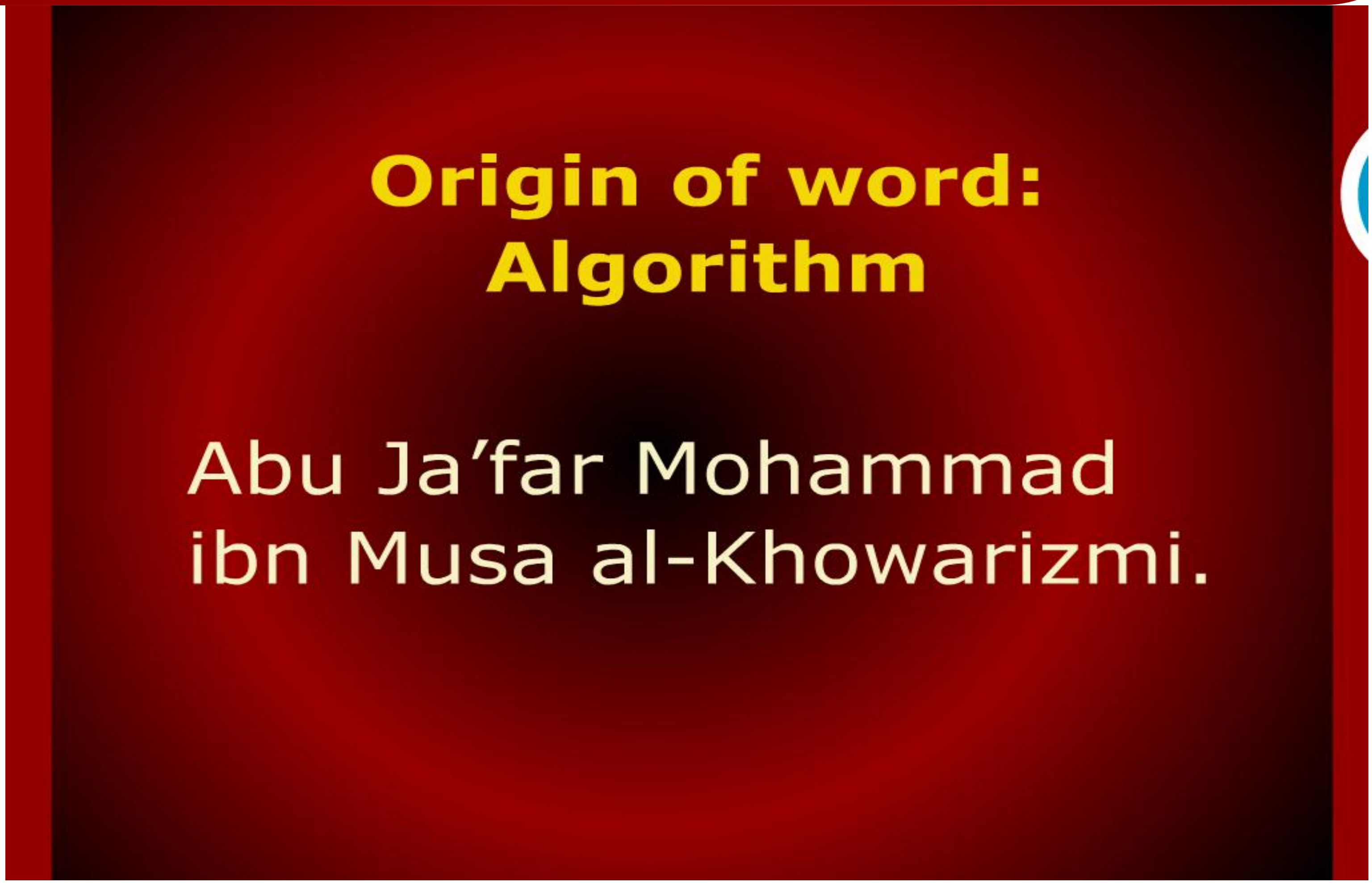
- *Textbook:* Introduction to Algorithms,Cormen,Leiserson, Rivest, Stein, 2nd edition, 2001 The MIT Press.
- Grading:
  - homework and projects
  - mid term exam
  - final

Origin of Word: ALGORITHM



Origin of Word: ALGORITHM

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al Kitab al-mukhatasar fi hisab al-jabr wa'l-muqabalah

Origin of Word: ALGORITHM

Origin of word: *Algorithm*

- It is from the titles of these writings and his name that the words algebra and algorithm are derived.
- As a result of his work, al-Khwarizmi is regarded as the most outstanding mathematician of his time

ALGORITHM informal definition

Algorithm: Informal Definition

- An algorithm is any well-defined computational procedure that takes some values, or set of values, as input and produces some value, or set of values, asb output.
- An algorithm is thus a sequence of computational steps that transform the input into output.

ALGORITHM, programming

Algorithms, Programming

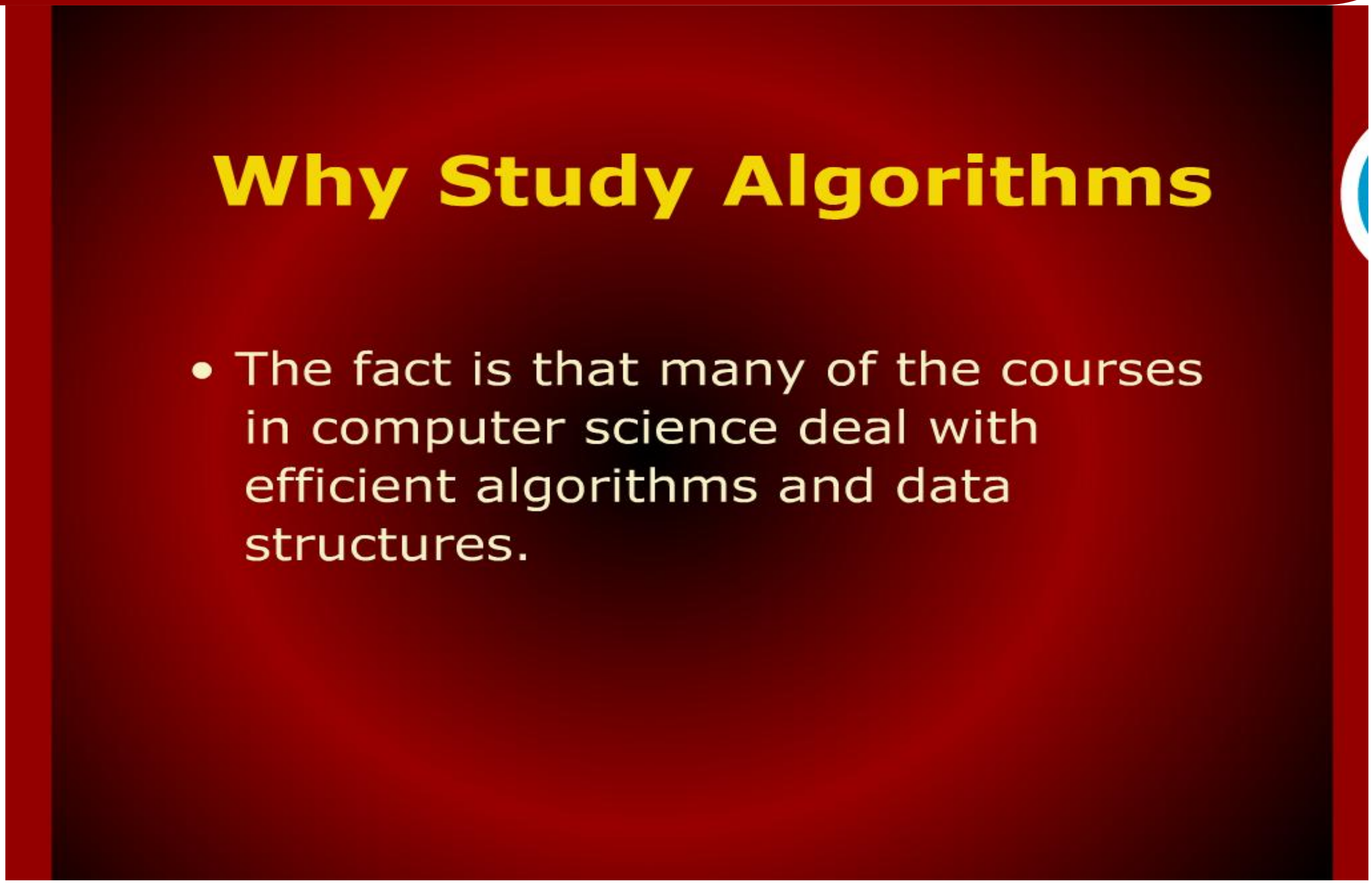
- A good understanding of algorithms is essential for *programming.*
- However, unlike a program, an algorithm is a *mathematical entity*

ALGORITHM, programming

Algorithms, Programming

- Algorithm is independent of a specific programming language, machine, or compiler.
- Algorithm design is about mathematical theory behind the design of good programs.

Why Study?


Why Study Algorithms
• The fact is that many of the courses in computer science deal with efficient algorithms and data structures.

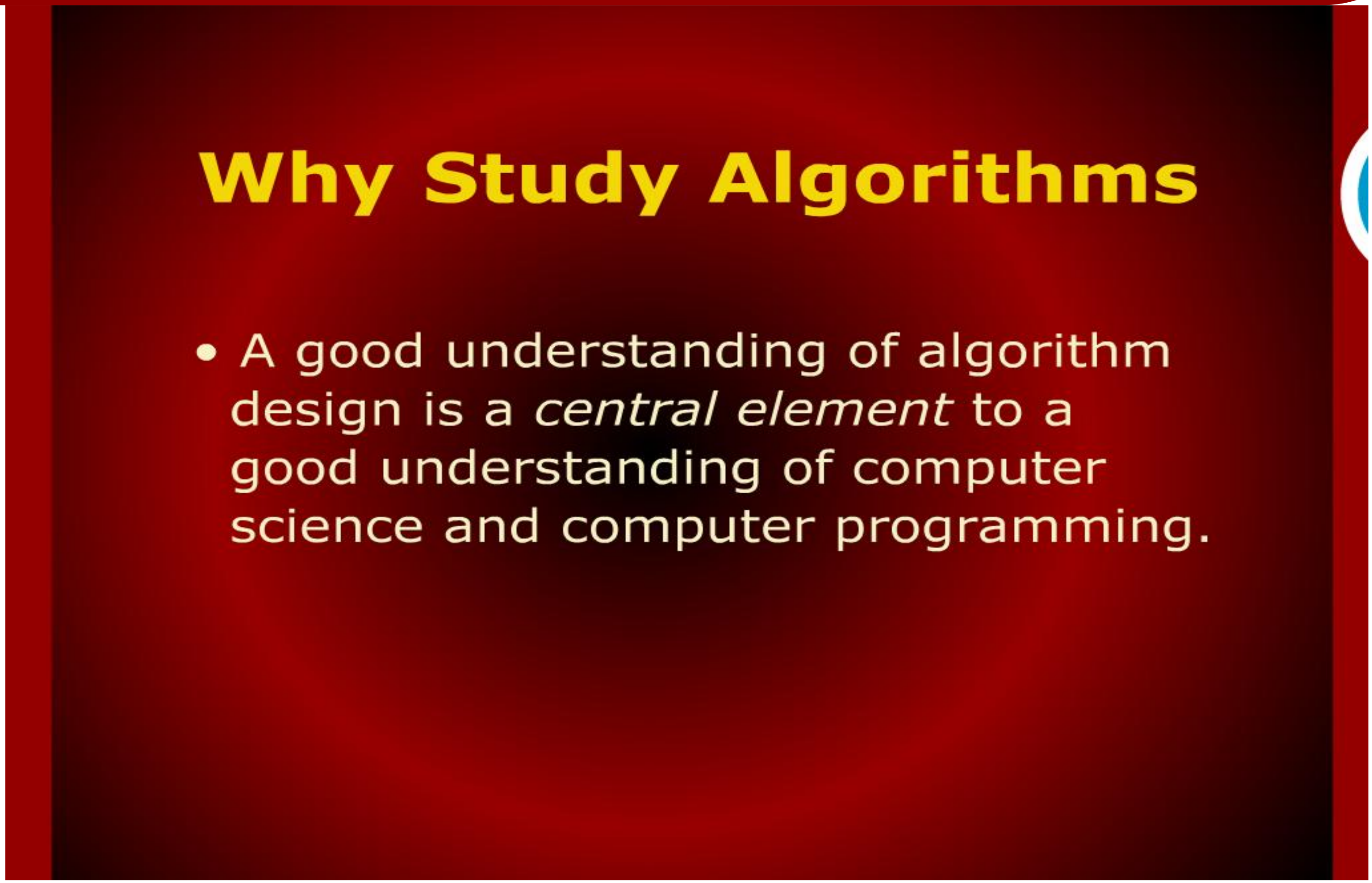
Why Study?

Why Study Algorithms

They apply these to various applications:

- compilers,
- operating systems,
- databases,
- artificial intelligence,
- computer graphics and vision,
- networks, etc.

Why Study?


Why Study Algorithms
• A good understanding of algorithm design is a *central element* to a good understanding of computer science and computer programming.

Course outline

Course Outline

This course will consist of four major sections.

1. Mathematical tools necessary for the analysis of algorithms. This will focus on asymptotics, summations, recurrences.

Course outline

Course Outline

2. Sorting: this will focus on different strategies for sorting, and use this problem as a case-study in different techniques for designing and analyzing algorithms.

Course outline

Course Outline

3. Collection of various algorithmic problems and solution techniques: Dynamic programming, greedy strategy, graphs.

Course outline

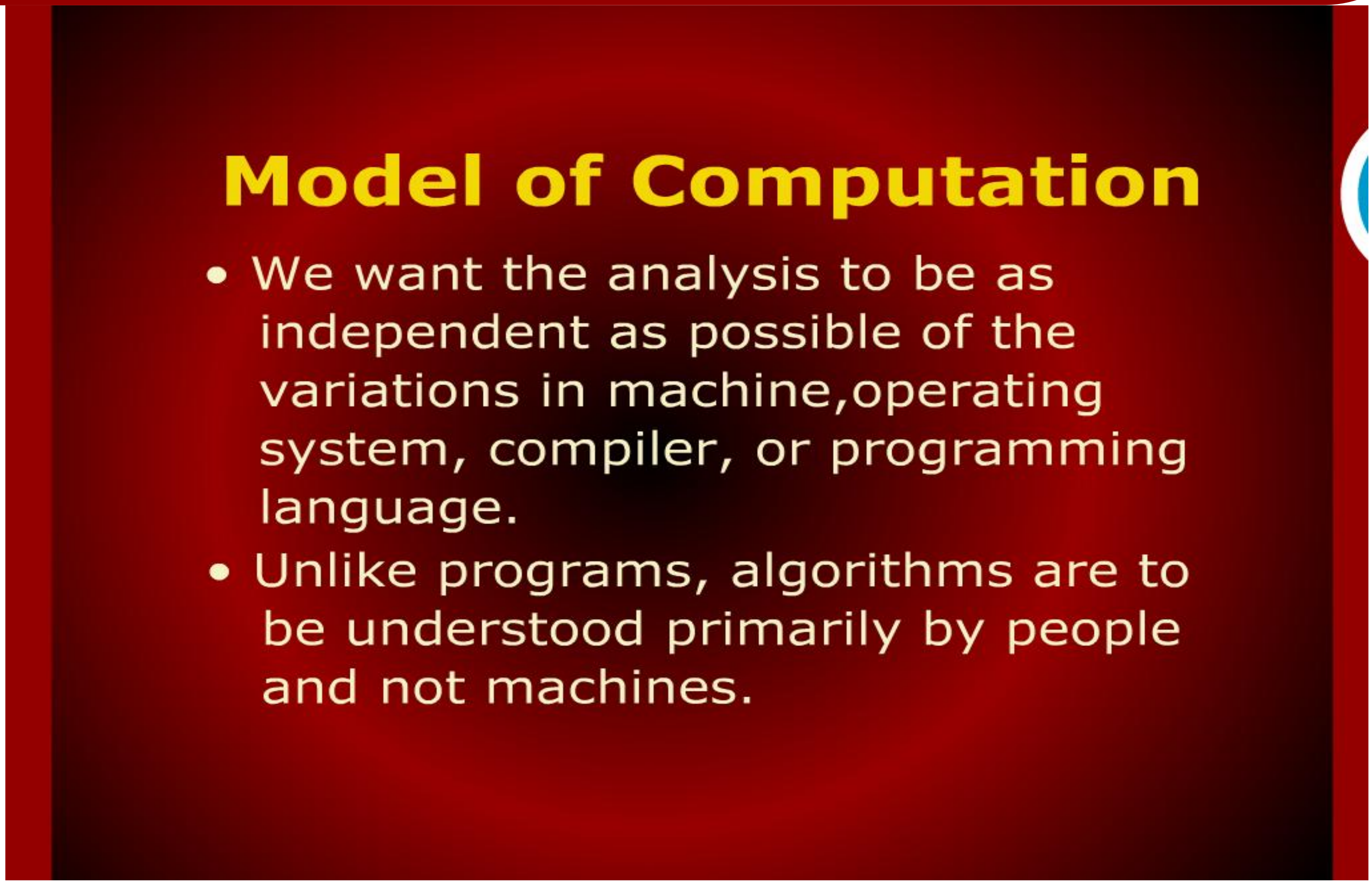
Course Outline

4. Introduction to the theory of NP-completeness.
   NP-complete problem are those for which no efficient algorithms are known, but no one knows for sure whether efficient solutions might exist.

Criteria for analyzing Algorithms

Criteria for analyzing Algorithms

Model of Computation



Model of Computation

Model of Computation

- Thus gives us quite a bit of flexibility in how we present our algorithms, and many low-level details may be omitted

Model of Computation

Model of Computation

- Ideally this model should be a reasonable abstraction of a standard generic single-processor machine.
- We call this model a *random access machine* or *RAM*.

Random Access Machine

Random Access Machine

- A RAM is an idealized machine with an infinitely large random-access memory.
- Instructions are executed one-by-one (there is no parallelism).
- Each instruction involves performing some basic operation on two values in the machine's memory

Basic RAM Operations

Basic RAM Operations

Basic operations include

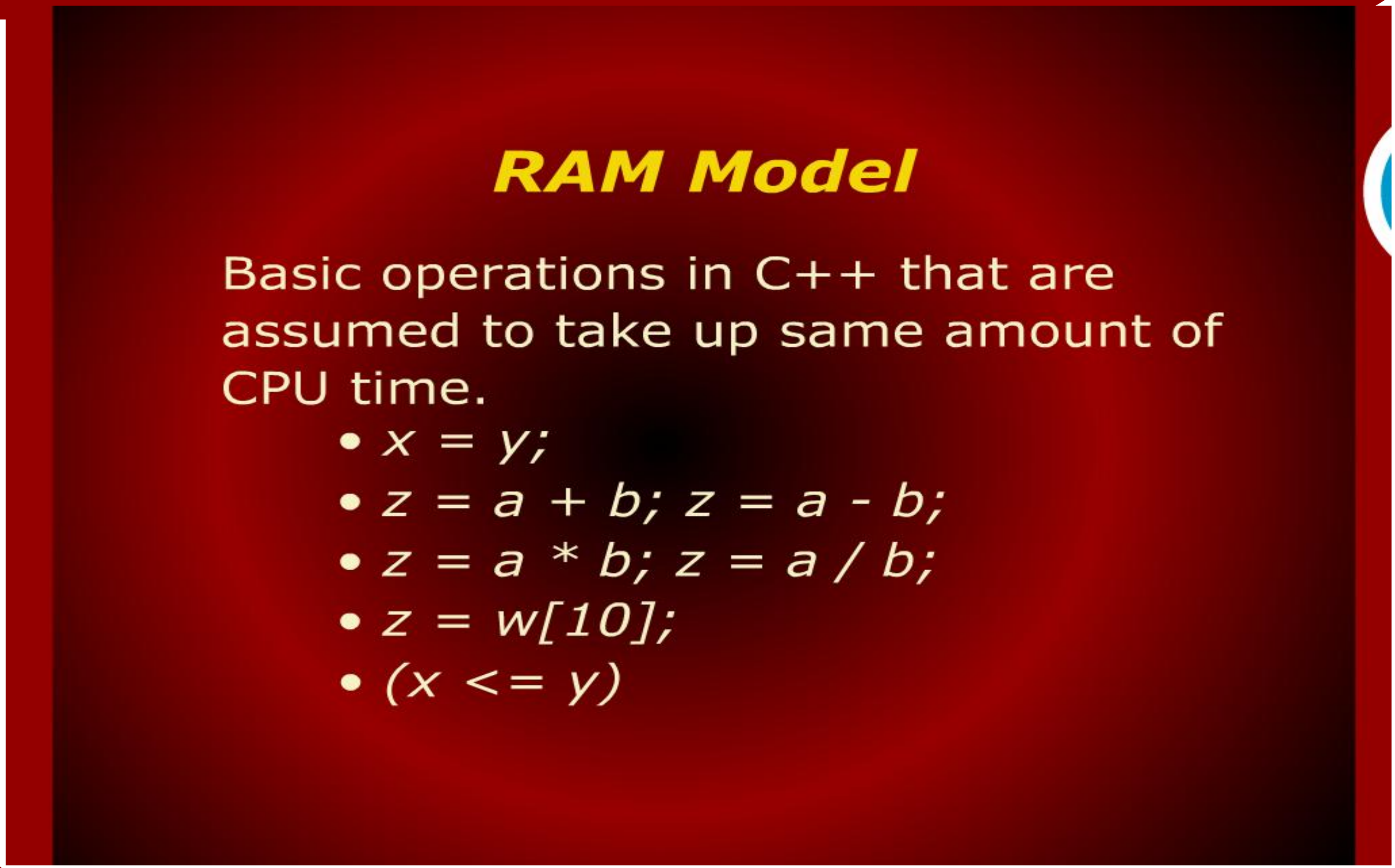
- assigning a value to a variable,
- computing any basic arithmetic operation (+, - , × and division) on integer values of any size
- performing any comparison or boolean operations
- accessing an element of an array (e.g. A[10])

RAM Model

RAM Model

- We assume that each basic operation takes the same constant time to execute.
- The RAM model does a good job of describing the computational power of most modern(nonparallel) machines.

RAM Model


RAM Model
Basic operations in C++ that are assumed to take up same amount of CPU time.
• x = y;
• z = a + b; z = a - b;
• z = a * b; z = a / b;
• z = w[10];
• (x <= y)